

The symmetric cube and quartic: tensor decompositions, Euler determinations, and reflection transport

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Abstract

Rungs 4–5 of the benchmark ladder at the layer the transport-closure calculus reaches exactly, for a level-one holomorphic Hecke eigenform with no temperedness input. The two tensor laws $\text{std} \otimes \text{Sym}^2 = \text{Sym}^3 \oplus \text{std}$ and $\text{Sym}^2 \otimes \text{Sym}^2 = \text{Sym}^4 \oplus \text{Sym}^2 \oplus \mathbf{1}$ are proven as finite weight-multiset identities and assembled globally: the tensor banks factor as Dirichlet convolutions of the compiled Sym^r banks. Every Sym^r bank is shown L-series-summable beyond an explicit polynomial threshold, whence the Euler determinations $L(\text{std} \otimes \text{Sym}^2, s) = L(\text{Sym}^3, s)L(\text{std}, s)$ and $L(\text{Sym}^2 \otimes \text{Sym}^2, s) = L(\text{Sym}^4, s)L(\text{Sym}^2, s)\zeta(s)$ hold on explicit half-planes: the symmetric cube and quartic L-series are the cofactors of compiled objects inside compiled tensor banks. A division-free transport lemma carries a self-dual reflection of the completed tensor object to the cofactor, locating each rung’s remaining analytic obligation in exactly one typed slot — the same slot the $r = 2$ rung’s lattice route discharged. Everything stated is proven unconditionally and machine-checked.

1 The tensor laws

Fix an eigenform seed with Satake parameters α_p and recall the clock systems $\text{Sym}^r : (\alpha^r, \alpha^{r-2}, \dots, \alpha^{-r})$.

Theorem 1 ($\text{std} \otimes \text{Sym}^2 = \text{Sym}^3 \oplus \text{std}$). *The six pairwise products of the standard clock (α, α^{-1}) with the Sym^2 clock $(\alpha^2, 1, \alpha^{-2})$ form the multiset $\{\alpha^3, \alpha, \alpha^{-1}, \alpha^{-3}\} \sqcup \{\alpha, \alpha^{-1}\}$: the Sym^3 clock together with the standard clock. (*tensor23_products*, *tensor23_decomposition* via the explicit relabeling *split23*.)*

Theorem 2 ($\text{Sym}^2 \otimes \text{Sym}^2 = \text{Sym}^4 \oplus \text{Sym}^2 \oplus \mathbf{1}$). *The nine pairwise products of two Sym^2 clocks form $\{\alpha^4, \alpha^2, 1, \alpha^{-2}, \alpha^{-4}\} \sqcup \{\alpha^2, 1, \alpha^{-2}\} \sqcup \{1\}$. (*tensor33_products*, *tensor33_decomposition* via *split33*.)*

Both are finite identities in $\alpha^{\pm 1}$, verified channel by channel.

*Machine-checked companion: `RequestProject/SymCubeQuartic.lean`, 14 named declarations, each at axiom footprint `{propext, Classical.choice, Quot.sound}` or a subset, no `sorry`, no `axiom`, zero `sorryAx`.

2 Global assembly

Through the closure law (banks of direct sums convolve) and the rank-uniform clock-bank identification (`bankArithmetic_symClock` — the compiled all-ranks join), the weight-level laws become bank-level factorizations:

Theorem 3 (tensor banks factor). $c_{\text{std} \otimes \text{Sym}^2} = c_{\text{Sym}^3} \star c_{\text{std}}$ and $c_{\text{Sym}^2 \otimes \text{Sym}^2} = c_{\text{Sym}^4} \star c_{\text{Sym}^2} \star \zeta$, as multiplicative arithmetic functions. (*tensor23_bank, tensor33_bank.*)

3 Summability and the Euler determinations

Hecke’s coefficient bound alone gives a single-power Satake estimate $\|\alpha_p\|, \|\alpha_p^{-1}\| \leq p^E$ uniformly in p (`satake_single_bound`), whence every clock channel obeys $\|\alpha_p^{r-2i}\| \leq p^{Er}$ (`symClock_norm_bound`) and the rank- r clock system is a polynomial Satake pair at every rank (`symrSatakePair`). The generic coefficient machinery then bounds the bank, $\|c_{\text{Sym}^r}(n)\| \leq n^{B_r}$ with $B_r = (r+1) + Er$ explicit, and:

Theorem 4 (summability at every rank). $L(c_{\text{Sym}^r}, s)$ converges absolutely for $\Re s > B_r + 1$ — with no temperedness input. (*symrBank_LSeriesSummable.*)

Theorem 5 (Euler determinations). On the common half-plane of absolute convergence,

$$L(\text{std} \otimes \text{Sym}^2, s) = L(\text{Sym}^3, s) L(\text{std}, s), \quad L(\text{Sym}^2 \otimes \text{Sym}^2, s) = L(\text{Sym}^4, s) L(\text{Sym}^2, s) \zeta(s).$$

The symmetric cube and quartic L -series are thereby determined as the cofactors of compiled objects inside the compiled tensor banks. (*sym3_euler_determination, sym4_euler_determination.*)

4 Reflection transport, and where the wall sits

Lemma 6 (division-free transport). If $T = Z \cdot Q$ pointwise and $T(1-s) = T(s)$ for all s , then $Z(1-s)Q(1-s) = Z(s)Q(s)$ for all s — the cofactor’s functional equation in cross-multiplied form, no division, no zero-set condition. (*FE_transport.*)

Applied with T the completed tensor object and Z the compiled cofactor ($L(\text{std})$ -completed for the cube; $\zeta \cdot L(\text{Sym}^2)$ -completed for the quartic, both of whose functional equations are compiled — the seed’s and the symmetric-square companion’s), Lemma 6 shows: *the entire remaining analytic content of rung 4 (resp. 5) is one self-dual reflection for the explicitly presented degree-8 (resp. degree-9) tensor bank.* That is the same typed slot which, at $r = 2$, the lattice route discharged (the symmetric-square companion); the classical discharge at $r = 3, 4$ is Kim-Shahidi [1], consumed at published strength where cited.

Addendum — the collapse compiled. Mining the corpus dissolved the composite walls to their irreducible cores: the standard rung of `CPSGenuineGL2StandardRung3D` derives its reflection from Hecke’s inversion (the modular slash law), and its Euler certificate is discharged at the eigenform seed (`SeedStandardRung.seedRungData` — the first inhabitant), giving the compiled completed Hecke equation $\Lambda(1-s) = (-1)^n \Lambda(s)$ with Λ entire and equal to $(s + (n - \frac{1}{2})) D(s)$ on the initial half-plane (`seed_selfdual FE, seed_symm_Lambda_eq`). With this and the compiled $\Lambda_\zeta \cdot \bar{\Lambda}$ reflections, the generic scalar-cancellation equivalence (`wall_collapse`) proves: the degree-6 tensor equation is *equivalent* to the Sym^3 single-object self-duality, and the degree-9

tensor equation to the Sym^4 one (`sym3_wall_collapse`, `sym4_wall_collapse`) — every cofactor reflection now a compiled theorem, the wall reduced to exactly one self-dual identity per rung with nothing else left standing.

Remark 7 (register and falsifiers). *Proven and machine-checked, unconditionally: both tensor laws, both bank factorizations, summability at every rank from Hecke’s bound alone, both Euler determinations, and the transport lemma. Cited, not re-proved: the classical analytic continuation of the tensor objects (Kim–Shahidi), which occupies exactly the one typed slot per rung identified above. A counterexample would be a prime at which the six (resp. nine) products fail the stated splits, a rank at which the bank escapes its polynomial bound, or a point of the common half-plane violating either Euler determination — each formally stated, so a counterexample would require a kernel failure.*

References

- [1] H. Kim and F. Shahidi, *Functorial products for $\text{GL}(2) \times \text{GL}(3)$ and the symmetric cube for $\text{GL}(2)$* , Ann. of Math. **155** (2002), 837–893; H. Kim, *Functoriality for the exterior square of GL_4 and the symmetric fourth of GL_2* , J. Amer. Math. Soc. **16** (2003), 139–183.
- [2] S. Gelbart and H. Jacquet, *A relation between automorphic representations of $\text{GL}(2)$ and $\text{GL}(3)$* , Ann. Sci. ÉNS **11** (1978), 471–542.